

Persuading a privately informed and inattentive receiver

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Grass workshop

Inattention and persuasion

Information does not come for free.

Information providers attract attention, and induce a course of action.

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1. Product advertisement;
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1. Product advertisement;
2. Regime censorship.

This paper studies persuasion of a **receiver** who is privately informed about the costs and benefits of information.

Overview

This paper studies persuasion à la Kamenica and Gentzkow when **Receiver** is privately informed and rationally inattentive.

Which types of Receiver pay attention? Does the possibility of screening help Sender?

Results:

1. Persuasion mechanisms are equivalent to experiments;
2. Upper censorships are optimal in applications;
3. Persuasion mechanisms are without loss under deterministic coordination mechanisms, but with loss under stochastic coordination mechanisms.

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Plan:

1. Model;
2. Literature;
3. Equivalence between persuasion mechanisms and experiments;
4. Optimal experiment with single-peaked outside option;
5. Generalized principal-agent setting (Myerson, 1982).

Model

Model

Sender and Receiver.

Payoff to Receiver from **action** $a \in \{0, 1\}$ and **effort** $e \in [0, 1]$ is

$$U_R(a, \theta, e, c, \lambda) = \underbrace{a(\theta - c)}_{\text{material payoff } u_R} - \underbrace{\lambda k(e)}_{\text{effort cost}},$$

in which:

$\theta \in [0, 1]$ is the state, $\sim$ continuous F_0 , with mean x_0 ;

$c \in [0, 1]$ is the outside option;

$\lambda \geq 0$ is the attention cost;

$k: [0, 1] \rightarrow \mathbb{R}_+$ is strictly convex (part of this talk).

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Receiver is privately informed about her **type**: $(c, \lambda) \in T$, independent of θ , drawn from H .

Persuasion mechanisms

A signal is $\sigma: \Theta \rightarrow \Delta M$.

A persuasion mechanism is a list of signals, $(\sigma_r)_{r \in T}$.

Timing

1. Sender commits to a persuasion mechanism $(\sigma_r)_{r \in T}$;
2. Receiver chooses a signal σ_r and effort e , knowing her type;
3. Nature draws θ according to F_0 ;
With probability e , Receiver observes a message m drawn from $\sigma_r(\theta)$;
4. Receiver chooses an action a .

Literature

Bayesian persuasion

Kamenica and Gentzkow (2011)... [Kolotilin et al. \(2017\)](#), Kolotilin (2018), Guo and Shmaya (2019), Kolotilin et al. (2022).

↑ rational inattention

Persuasion of an inattentive receiver

[Bloedel and Segal \(2021\)](#), [Wei \(2021\)](#).

↑ private information, inflexible inattention

Persuasion with “outside information”

Bizzotto et al. (2021); Dworczak and Pavan (2022); Matysková and Montes (2023).

Dynamic information provision

Knoepfle (2024); Liao (2021); Jain and Whitmeyer (2025); Che et al. (2023); Au and Whitmeyer (2023); Hébert and Zhong (2025)...

Attention management

Lipnowski et al. (2020).

**Equivalence between persuasion
mechanisms and signals**

Mean distribution

Optimal action given belief μ of type (c, λ) Receiver is:

$$a^* \in [\mathbf{1}\{\mathbb{E}_\mu[\theta] > c\}, \mathbf{1}\{\mathbb{E}_\mu[\theta] \geq c\}].$$

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Marginal benefit of effort given belief distribution p of type- (c, λ)

Receiver is:

$$A(c) = \underbrace{\mathbb{E}_p[\mathbb{E}_\mu[u_R(a^*, c, \theta)]]}_{\text{value of informed action}} - \underbrace{u_R(\mathbf{1}\{x_0 \geq c\}, c, x_0)}_{\text{value of uninformed action}}.$$

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The mean distribution of any experiment describes equilibrium action and effort.

Signals as information policies

We identify a signal with its **information policy**.

The *information policy* of the CDF F is the function I_F given by

$$I_F(x) = \int_0^x F(\tilde{x}) \, d\tilde{x}.$$

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(1) I_F is convex;

(2) $I_{\bar{F}} \leq I_F \leq I_{F_0}$, in which:

$\bar{F}$ is the CDF induced by an **uninformative signal**.

F_0 is the CDF induced by a **fully informative signal**.

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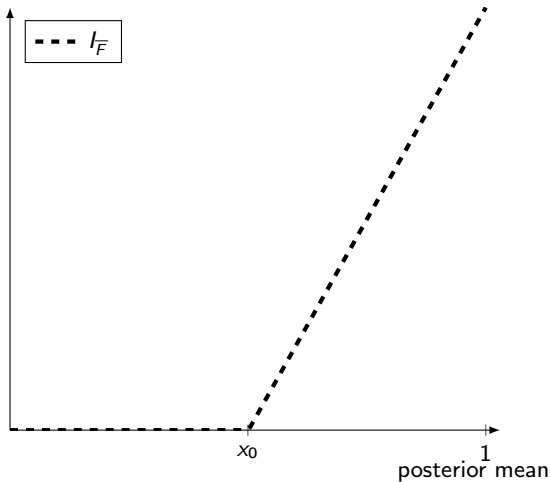
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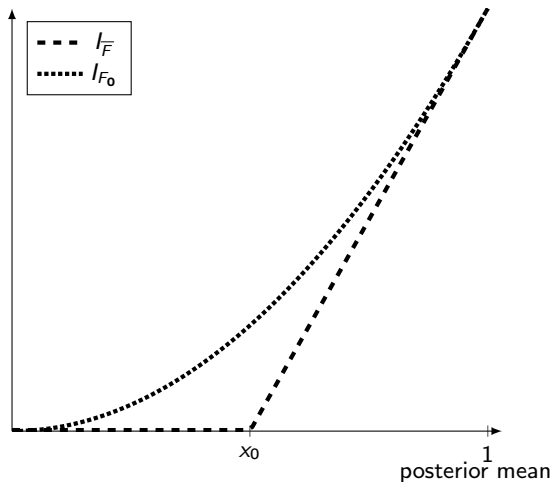
Lemma 1 (Gentzkow and Kamenica 2016; Kolotilin 2018.)

If $I: \mathbb{R}_+ \rightarrow \mathbb{R}_+$ satisfies (1) and (2), then there exists a signal that induces a posterior-mean CDF with information policy I .

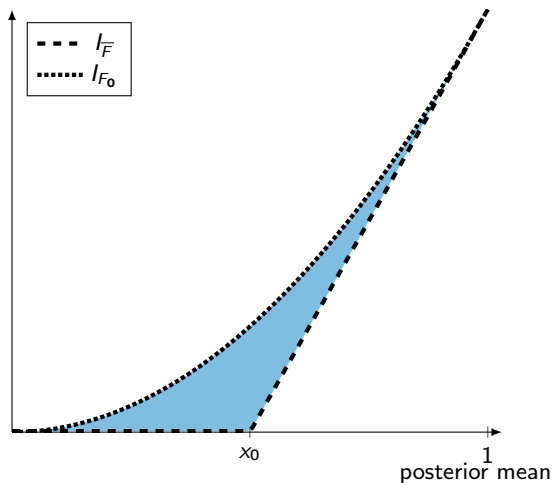
Information policies



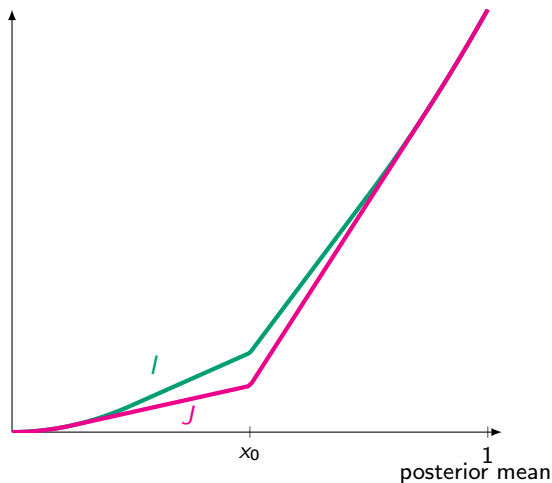
Information policies



Information policies



Information policies



$I \geq J$ only if I is “more informative” than J .

Interval structure of the extensive margin

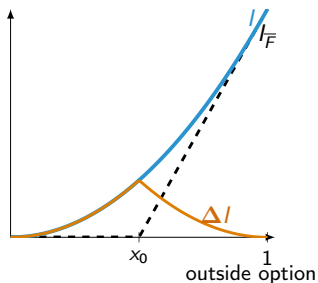
The *net informativeness* of I is denoted by

$$\Delta I = I - I_{\bar{F}}.$$

Lemma 2

The marginal benefit of effort given information policy I satisfies, for all c ,

$$A(c) = \Delta I(c).$$



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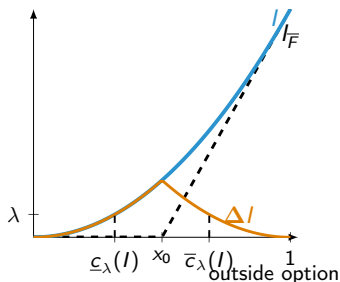
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Lemma 3

The marginal benefit of effort given information policy I satisfies, for all c ,

$$A(c) = \Delta I(c).$$



Linear Cost ($k(e) = e$).
There exist *cutoff types*,
given attention cost λ :
 $c_{\lambda}(I)$, $\bar{c}_{\lambda}(I)$.

Persuasion mechanisms

The *value of information policy* I is the interim payoff of type- (c, λ)
Receiver:

$$V_R(\Delta I(c), \lambda) := \max_{e \in [0,1]} \underbrace{e \Delta I(c)}_{\text{increasing differences}} - \lambda k(e).$$

- Information and attention are complements.

A *persuasion mechanism* is a menu of information policies: $(I_r)_{r \in T}$.

A persuasion mechanism $(I_r)_{r \in T}$ is *incentive compatible* (IC) if, for all types $(c, \lambda) \in T$ and reports $\widetilde{(c, \lambda)} \in T$,

$$V_R(\Delta I_{(c, \lambda)}(c), \lambda) \geq V_R(\Delta I_{\widetilde{(c, \lambda)}}(c), \lambda).$$

Persuasion mechanisms

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► Information and attention are complements.

An IC persuasion mechanism $(I_r)_{r \in T}$ is *equivalent to information policy* J if, for every type (c, λ) , the optimal effort and action are the same:

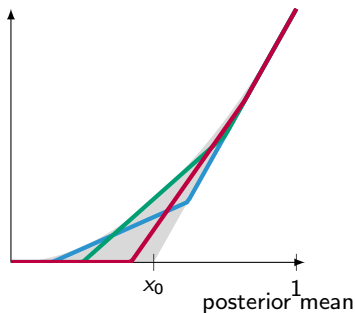
- $\arg\max_{e \in [0,1]} e \Delta I_{(c,\lambda)}(c) - \lambda k(e) \subseteq \arg\max_{e \in [0,1]} e \Delta J(c) - \lambda k(e)$
- if $(0, 1] \cap \arg\max_{e \in [0,1]} e \Delta I_{(c,\lambda)}(c) - \lambda k(e) \neq \emptyset$, then $\partial I_{(c,\lambda)}(c) \subseteq \underbrace{\partial J(c)}_{\substack{[J'_-(c), J'_+(c)] \\ \text{subdifferential}}}$

Equivalence

Proposition 1

For every IC persuasion mechanism, there exists an equivalent information policy.

Proof sketch



1. By ET: (c, λ) 's rent from report r is $\int_0^{\Delta I_r(c)} \operatorname{argmax}_e \{ev - \lambda k(e)\} dv$.
 - An active report at c maximizes (c, λ) 's rent.

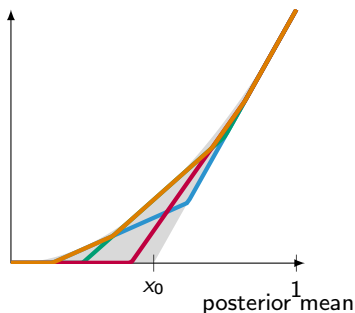
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$$\text{— } x \mapsto \sup_{r \in T} I_r(x)$$



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 - An *active* report at c maximizes (c, λ) 's rent.
2. The sup information policy has an inclusion-larger subdifferential than I_r for active r .

Optimal information policy

Sender's problem

The Sender's expected payoff from information policy I is

$$V_S(I) = \int_T \underbrace{e^*(I, c, \lambda)}_{\text{extensive margin}} \underbrace{(1 - I'_-(c) - \mathbf{1}\{x_0 \geq c\})}_{\text{intensive margin}} + \mathbf{1}\{x_0 \geq c\} dH(c, \lambda).$$

Assumption: (1) str. convex k ; (2) Receiver chooses action 1 if indifferent.

I is an optimal information policy if it solves the Sender's problem:

$$\sup_{I: \mathbb{R}_+ \rightarrow \mathbb{R}_+} V_S(I) \text{ subject to: } I \text{ is convex, } I_{\overline{F}} \leq I \leq I_{F_0}.$$

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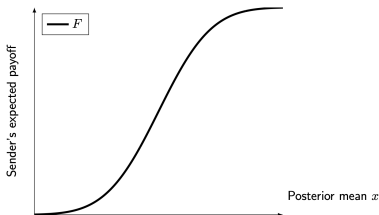
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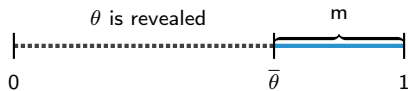
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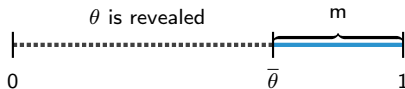
Assumption (SPness): the outside option c admits a single-peaked and abs. cont. PDF.



Optimal signal



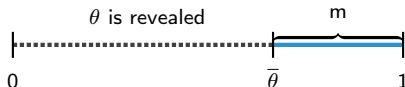
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Proposition 2

Under the SPness assumption, there exists an optimal information policy that is induced by an upper censorship signal.

Optimal signal



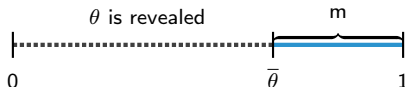
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Proof sketch

- ▶ By increasing differences, the information policy $x \mapsto e^* I(x) + (1 - e^*) I_{\bar{F}}(x)$ is nondecreasing in I ;
- ▶ By SPness,

$$V_S(I) = \int_{\lambda=0}^1 \int_{c=0}^1 V_R(\Delta I(c), \lambda) h'_c(c) \, dc h_\lambda(\lambda) \, d\lambda$$

Application to media censorship

- (1) Sender knows the attention cost λ .
- (2) k is linear.
- (3) Sender's payoff is:

$$U_S(a, \mathbf{e}) = \psi a + \gamma \mathbf{e}.$$

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- ▶ $\psi \geq 0$ is the mobilizing character of the government.
- ▶ $\gamma \geq 0$ is the size of the media market.

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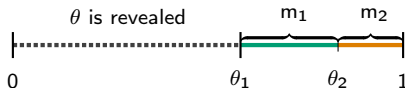
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Proposition 3

Under (1), (2), (3), and SPness, there exists an optimal information policy that is induced by a bi-upper censorship signal.



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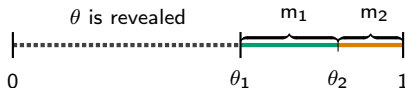
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Proposition 3

Under (1), (2), (3), and SPness, there exists an optimal information policy that is induced by a bi-upper censorship signal.



Kolotilin et al. (2022): no media market; Gehlbach and Sonin (2014): Sender knows c , and does not know λ .

Coordination mechanisms

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A *coordination mechanism* is a triple (R, M, π) in which

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- ▶ A coordination mechanism (R, M, π) is *direct* if: $R = T$, $M = [0, 1]$;
- ▶ A direct coordination mechanism π is IC if:
 1. truthful reporting;
 2. obeying the effort recommendation;are optimal for Receiver.

Proposition 4 (Myerson 1982.)

For every coordination mechanism G , for every equilibrium Sender's payoff v in G , there exists a direct and IC coordination mechanism π such that

$$\underbrace{V(\pi)}_{\text{Sender's payoff in the truthful \& obedient eq'm of } \pi} = v.$$

Coordination mechanisms

- ▶ A coordination mechanism (R, M, π) is *deterministic* if there exists $(I_r)_{r \in T}$ such that, for all r , $\pi(\{I_r\} \times M | r) = 1$.
- ▶ An IC persuasion mechanism $(I_r)_{r \in T}$ induces a Sender's payoff v if:

$$v = \int_T e^*(I_{(c,\lambda)}, c, \lambda)(1 - (I_{(c,\lambda)})'_-(c) - \mathbf{1}\{x_0 \geq c\}) + \mathbf{1}\{x_0 \geq c\} dH(c, \lambda).$$

Proposition 5

For every deterministic coordination mechanism G , for every equilibrium Sender's payoff v in G , there exists a persuasion mechanism that induces the Sender's payoff v .

Correlating recommendation and information

Example: Linear k , uniform θ , $c = 3/4$ known to Sender, and two types:

Low ($\lambda_1 \approx 0.01$), Prob = $8/9$;

High ($\lambda_2 = 0.02$), Prob = $1/9$.

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The **optimal deterministic coordination mechanism** may:

1. Induce effort from both with highly informative J_H ;
2. Exclude high-cost type with less informative J_L .

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The **optimal deterministic coordination mechanism** may:

1. Induce effort from both with highly informative J_H ;
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Both cases yield a maximum payoff of 0.4, whereas the **coordination mechanism** such that

Low gets a deterministic recommendation, 1, and some information I_L .

High gets a **lottery**:

- ▶ Prob $p \approx 0.3$: Recommend $e = 1$ and the informative J_H ;
- ▶ Prob $1 - p \approx 0.7$: Recommend $e = 0$ and no information;

improves upon 0.4.

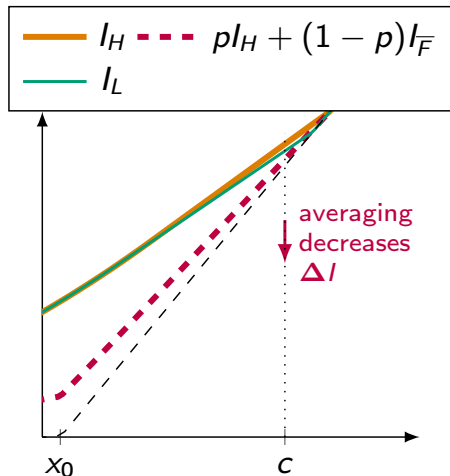
Correlating recommendation and information

Example: Linear k , uniform θ , $c = 3/4$ known to Sender, and two types:

Low ($\lambda_1 \approx 0.01$), Prob = $8/9$;

High ($\lambda_2 = 0.02$), Prob = $1/9$.

- ▶ High (weakly) prefers the lottery because ΔI_L does not cover λ ;
- ▶ Low prefers I_L and effort 1 with probability 1 to the lottery;



Alternative approaches to inattention

Cost of learning about the state (Wei, 2021):

- ▶ Costly reduction of uncertainty about the state;
- ▶ Flexible information acquisition about θ ;

Interpretation: **Delegated learning** (e.g., analyzing a dataset).

Cost of processing the message (Bloedel and Segal, 2021):

- ▶ Costly entropy reduction about the message;
- ▶ Flexible information acquisition about m ;

Interpretation: **Learning from communication** (e.g., decoding ads).

Cost of exposure:

- ▶ Cost $\lambda k(e)$ is independent of the informativeness;
- ▶ Receiver chooses mixtures of full and null observation;

Interpretation: **Learning via exposure** (e.g., scrolling feed, platform fee).

Conclusion

- ▶ With inattention and private information:
 1. Persuasion mechanisms are equivalent to signals;
 2. Upper censorships are optimal;
 3. Persuasion mechanisms are without loss under deterministic coordination mechanisms;
 4. Stochastic coordination mechanisms can correlate the signal with the attention recommendation.
- ▶ In the paper:
 1. Comparative statics wrt λ ;
 2. Persuasion mechanisms are without loss if c is known and k is isoelastic.

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Thanks!

Extra slides

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Setting

Let $X := [0, 1]$ be the state space, and F_0 a continuous prior CDF on X with mean $x_0 \in (0, 1)$.

Definition 1 (Information Policies)

Let $\mathcal{C}(X)$ be the space of real-valued continuous functions on X endowed with the supremum norm $|\cdot|_\infty$. The uninformative policy is $\underline{I}(x) := (x - x_0)_+$, and the fully informative policy is $I_0(x) := \int_0^x F_0(t) dt$. The set of feasible information policies is

$$\mathcal{I} := \{I \in \mathcal{C}(X) \mid I \text{ is convex, } \underline{I} \leq I \leq I_0, \partial I(X) \subseteq [0, 1]\}.$$

For any $I \in \mathcal{I}$, let I' denote its right-derivative.

Lemma 4 (Lemma 1)

The mapping $T: F \mapsto (x \mapsto \int_0^x F(y) dy)$ is a topological homeomorphism from the space of Bayes-plausible posterior-mean CDFs under the L^1 norm to the metric space $\langle \mathcal{I}, |\cdot|_\infty \rangle$.

Equilibrium

Let $T := X \times \mathbb{R}_+$ be the type space endowed with a probability measure P . Given $I \in \mathcal{I}$ and effort $e \in [0, 1]$, define the effective posterior-mean CDF $(e \odot I') := eI' + (1 - e)\bar{F}$, where $\bar{F}(x) := \mathbf{1}\{x \geq x_0\}$.

Definition 2 (Equilibrium)

An equilibrium is a tuple $\langle I^*, e, \alpha \rangle$, where $I^* \in \mathcal{I}$, $e: T \times \mathcal{I} \rightarrow [0, 1]$ is measurable, and $\alpha: T \times X \rightarrow [0, 1]$ is measurable, such that:

1. **Action Optimality:** For all $x \in X$ and $(c, \lambda) \in T$,

$$\alpha(c, \lambda, x) \in \operatorname{argmax}_{b \in [0, 1]} b(x - c).$$

2. **Effort Optimality:** For all $(c, \lambda) \in T$ and $\hat{I} \in \mathcal{I}$,

$$e(c, \lambda, \hat{I}) \in \operatorname{argmax}_{\tilde{e} \in [0, 1]} \int_X (x - c)_+ d(\tilde{e} \odot \hat{I}')(x) - \lambda k(\tilde{e}).$$

3. **Sender Rationality:** $I^* \in \operatorname{argmax}_{\hat{I} \in \mathcal{I}} \hat{W}(\hat{I}, e, \alpha)$, where

$$\hat{W}(\hat{I}, e, \alpha) := \int_T \int_X \alpha(c, \lambda, x) d(e(c, \lambda, \hat{I}) \odot \hat{I}')(x) dP(c, \lambda).$$

Receiver's Value and the Extensive Margin

Definition 3

The net informativeness operator is $\Delta: \mathcal{I} \rightarrow \mathcal{C}(X)$, defined by $\Delta I := I - \underline{I}$. The receiver's value function $V_\lambda: [0, 1] \times \mathbb{R}_+ \rightarrow \mathbb{R}$ is:

$$V_\lambda(e, \Delta I(c)) := e\Delta I(c) - \lambda k(e).$$

Let $E_\lambda(\Delta I(c)) := \operatorname{argmax}_{e \in [0, 1]} V_\lambda(e, \Delta I(c))$.

Lemma 5 (Lemmata 2 and 3)

The marginal benefit of effort $B(I, c) := \int_X (x - c)_+ dI'(x) - (x_0 - c)_+$ satisfies $B(I, c) = \Delta I(c)$. Furthermore:

1. $V_\lambda(e, \Delta I(c))$ exhibits strictly increasing differences in $(e, \Delta I(c))$.
2. Every selection from the correspondence $E_\lambda(\cdot)$ is nondecreasing.
3. Given an equilibrium $\langle I^*, e(\cdot), \alpha \rangle$ and $\lambda \geq 0$, the active set $\{c \in X \mid e(c, \lambda, I^*) > 0\}$ is an interval, and contains x_0 if non-empty.

Optimality of Upper Censorships

Assumption 1 (SPness)

For all $\lambda \geq 0$, the conditional distribution of c given λ admits an absolutely continuous density $g(\cdot \mid \lambda)$. There exists a peak $p \in (0, 1)$ such that, for all λ , $g(\cdot \mid \lambda)$ is nondecreasing on $[0, p]$ and nonincreasing on $[p, 1]$.

Definition 4

For any state $\bar{\theta} \in X$, the $\bar{\theta}$ -upper censorship $I_{\bar{\theta}} \in \mathcal{I}$ is the unique information policy satisfying:

$$I_{\bar{\theta}}(x) = \begin{cases} I_0(x), & \text{if } x \in [0, \bar{\theta}] \\ \max\left\{I_0(\bar{\theta}) + F_0(\bar{\theta})(x - \bar{\theta}), \underline{I}(x)\right\}, & \text{if } x \in (\bar{\theta}, 1]. \end{cases}$$

Proposition 6 (Proposition 2)

Let Assumption 1 hold. There exists an equilibrium $\langle I^*, e(\cdot), \alpha \rangle$ wherein I^* is a $\bar{\theta}$ -upper censorship. Moreover, for any two equilibria $\langle I_1^*, e_1, \alpha_1 \rangle$ and $\langle I_2^*, e_2, \alpha_2 \rangle$, $\hat{W}(I_1^*, e_1, \alpha_1) = \hat{W}(I_2^*, e_2, \alpha_2)$.

Application

Setup: Sender's utility is $U_S(a, e) = a + \gamma e$ for $\gamma \geq 0$. Cost is linear: $k(e) = e$. Let $V_S(I, e, \alpha)$ denote the Sender's ex-ante expected utility under U_S .

Definition 5

An information policy $I \in \mathcal{I}$ is a bi-upper censorship if there exist $0 \leq \theta_1 \leq x_1 \leq x_2 \leq 1$ and $m = \frac{I(x_2) - [l_0(\theta_1) + F_0(\theta_1)(x_1 - \theta_1)]}{x_2 - x_1}$ such that:

$$I(x) = \begin{cases} l_0(x), & \text{if } x \in [0, \theta_1] \\ l_0(\theta_1) + F_0(\theta_1)(x - \theta_1), & \text{if } x \in (\theta_1, x_1] \\ I(x_2) - m(x_2 - x), & \text{if } x \in (x_1, x_2] \\ \underline{I}(x), & \text{if } x \in (x_2, 1]. \end{cases}$$

Proposition 7 (Proposition 3)

Let $P(X \times \{\lambda^*\}) = 1$ for some $\lambda^* \geq 0$. Let Assumption 1 hold with $p \geq x_0$. For every equilibrium $\langle I^*, e(\cdot), \alpha \rangle$, there exists a bi-upper censorship $\hat{I} \in \mathcal{I}$ such that $V_S(\hat{I}, e, \alpha) \geq V_S(I^*, e, \alpha)$.

Coordination Mechanisms

Definition 6

A coordination mechanism is a tuple $G = (R, \mathcal{R}, M, \mathcal{M}, \pi)$ where $(R, \mathcal{R})$ is a measurable space of reports, $(M, \mathcal{M})$ is a measurable space of messages, and $\pi: R \rightarrow \Delta(\mathcal{I} \times M)$ is a Markov kernel. G is *direct* if $(R, \mathcal{R}) = (T, \mathcal{B}_T)$ and $(M, \mathcal{M}) = ([0, 1], \mathcal{B}_{[0,1]})$.

Definition 7

A participation strategy (ρ, δ) consists of measurable functions $\rho: T \rightarrow R$ and $\delta: T \times M \rightarrow [0, 1]$. It is an equilibrium in G if no type $t \in T$ prefers any measurable deviation $\eta: M \rightarrow [0, 1]$ upon reporting $r \in R$. A direct mechanism is Incentive Compatible (IC) if truth-telling and obedience $(\rho^*(t) = t, \delta^*(t, m) = m)$ forms an equilibrium.

Proposition 8 (Proposition 4)

For every coordination mechanism G and every equilibrium (ρ, δ) in G , there exists an IC direct coordination mechanism $F = (T, [0, 1], \pi^F)$ such that $V_0^F(\rho^*, \delta^*) = V_0^G(\rho, \delta)$, where V_0 denotes the Sender's ex-ante expected utility.

Deterministic Mechanisms

Definition 8

A coordination mechanism $G = (R, \mathcal{R}, M, \mathcal{M}, \pi)$ is *deterministic* if for all $r \in R$, the marginal distribution of $\pi(\cdot \mid r)$ on $\mathcal{I}$ is a Dirac measure.

An *extended persuasion mechanism* is a direct mechanism $G = (T, \mathcal{B}_T, M, \mathcal{M}, \pi)$ admitting a measurable function $f: T \rightarrow \mathcal{I}$ and a Markov kernel $\pi_M: T \rightarrow \Delta(M)$ such that for all $\bar{I} \in \mathcal{B}_\mathcal{I}$ and $\bar{M} \in \mathcal{M}$:

$$\pi(\bar{I} \times \bar{M} \mid t) = \pi_M(\bar{M} \mid t) \mathbf{1}\{f(t) \in \bar{I}\}.$$

Proposition 9 (Proposition 5)

For every deterministic coordination mechanism G and equilibrium (ρ, δ) in G , there exists an extended persuasion mechanism G' and an equilibrium (ρ', δ') in G' with $\rho' = \rho^*$ such that $V_0^{G'}(\rho', \delta') = V_0^G(\rho, \delta)$.